

Omar Alaa

Egypt | omaralaa4526@gmail.com | 01070145013 | LinkedIn | GitHub | Portfolio

Professional Summary

Results-driven Artificial Intelligence student (Expected 2027) with hands-on experience in Machine Learning, DevOps, and Embedded Systems. Proven ability to build end-to-end ML pipelines achieving up to 99.47% accuracy, alongside practical experience automating deployments with Jenkins and Docker. Adept at cross-functional collaboration, curriculum development, and passionate about deploying scalable, production-ready AI solutions.

Education

Badr University in Assiut (BUA)

Faculty of Artificial Intelligence and Data Science

Bachelor's Degree in Artificial Intelligence

- Relevant Coursework: Machine Learning, Digital Image Processing, Embedded Systems, Computer Architecture

Experience & Training

Python & DevOps Intern

Telecommunications Company

- Engineered automated CI/CD pipelines using Jenkins, reducing manual deployment and enforcing DevOps best practices
- Deployed and orchestrated Docker containers, standardizing application packaging and improving environment consistency
- Configured and administered Ubuntu Linux servers, optimizing environment setups and streamlining deployment workflows for the engineering team

Co-founder & Computer Science Instructor

Youth Education Initiative

- Co-founded an educational academy and developed a comprehensive "Level 0" programming curriculum for youth aged 12 to 19
- Instructed a class of [X] students in fundamental computer science concepts, fostering problem-solving skills and technical literacy
- Coordinated curriculum planning and delivery with a core team, ensuring an engaging and structured learning environment

Certifications

- AI Fundamentals Training – Information Technology Institute (ITI)
- Python Programming – MaharaTech
- Machine Learning Specialization
- NTI Training Programs

Projects

End-to-End ML Pipeline for Handwritten Digit

Recognition

- Built and deployed a CNN model using PyTorch achieving 99.47% accuracy on MNIST dataset
- Developed a Flask backend with image preprocessing (cropping, centering, padding) to improve prediction performance
- Containerized the application using Docker and implemented CI/CD pipeline using Jenkins and SonarQube

Automated Crop Disease Detection

- Implemented object detection system using YOLOv8 for crop disease classification, achieving 75.8% validation accuracy on the PlantDoc dataset
- Built data pipeline and trained model on PlantDoc dataset using Google Colab

- Performed inference testing to ensure model accuracy and reliability

E-commerce Web Application

- Developed scalable backend using Flask framework, supporting concurrent user sessions
- Designed and integrated PostgreSQL database for managing users and products
- Implemented secure data handling and backend logic for e-commerce operations

Embedded Systems & Arduino Projects

- Designed and implemented motor control systems using H-Bridge and PWM
- Built multiple embedded applications using Arduino microcontrollers, including obstacle-avoiding mechanics.

Machine Learning Capstone Project

- Implemented and compared multiple machine learning models including neural networks for classification tasks
- Analyzed model performance and optimized accuracy across different approaches

Computer Vision Project

- Applied image processing techniques including edge detection (Sobel, Prewitt) and filtering using OpenCV
- Performed bit-plane analysis and image enhancement operations

Skills

Programming: Python, C++ , Java, JavaScript

Web Development: HTML, CSS, Flask

Machine Learning & AI: PyTorch, YOLOv8, OpenCV, Neural Networks, Computer Vision, Deep Learning

DevOps: Docker, Jenkins, SonarQube

Databases: SQL

Tools: Git, GitHub, Arduino, Ubuntu Server